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Course:

**Sustainable Corporate Finance
Response Paper**

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Abstract

In an era when the world struggles with various sustainable challenges within various realms of fragmentation amongst nations, becoming progressively difficult to themselves have found means by which they can hope to spur some real progress on the ground toward environmental objectives. It is argued in this paper that a combination of industrial policies strategically defined so that they become "green" by major economies together with the enforcement of climate litigation and networked respect for the varying definitions of corporate purpose across jurisdictions provides the best-known second best solution for sustainability aims in the international domain. It outlines the limitations of a single, universally defined corporate purpose, emphasizing the potential for powerful actors to make definitions that suit their own agendas. Unilateral green industrial policies appear as a pragmatic alternative with policies like the US Inflation Reduction Act, EU Green Deal, and India's renewable energy targets being prime movers for clean technology development. The eminently distinctive cases within climate litigation lead way to corporate accountability on the basis of the violations of fundamental human rights, mostly in the context of demarcating government inactivity and making companies accountable for Scope 3 emissions.

Remote sensing data provide an urgent sense of environmental change that must be expedited. Statistical analysis on various indices on the temperature of the land surface, vegetation index, and land-use changes also tables a case for stronger policies and mitigation strategies. The pathway remains rough around the edges, but this second-best alternative is at least capable of taking the first steps of enabling a real undertaking under all the market dynamics, braving the geo-economic entanglement.

Keywords: sustainability, green industrial policy, climate litigation, corporate purpose, satellite imagery, environmental accountability

Introduction

Due to prevalent global sustainability challenges and zealous geopolitics, achieving meaningful progress in regards to the environment is all the harder. Absent that unifying compromise from the political factions, the focus is on other avenues that promise to realise at least some progress on sustainability objectives within the constraints of the current geo-economic reality. The writers make the case here for strategic green industrial policy implemented by major economies, robust climate litigation, and a respect for divergent approaches of corporate purpose varying by jurisdiction as a very feasible second-best way to advance sustainability objectives across the globe.

Limitations of Universally Defined Corporate Purpose

By their nature, academic definitions of corporate purpose tend to be varyingly narrow in scope, skewed by jurisdictional considerations that, for the most part, fall short of factoring in extraterritorial ramifications of corporate actions or the numerous legal frameworks that apply to them (Nurgozhayeva and Puchniak, 2020). As issues relating to climate change and sustainability defy national boundaries, a more universal understanding of corporate purpose has become necessary. However, a globally defined corporate purpose is not without perils.

When powerful states, firms, and organizations try to define corporate purpose at the global level, risk emerges that the definitions will serve these powerful players and their geostrategic interests at the expense of less privileged communities. China's approach serves as a case in point to show this chess game of moving pieces on the board. The EU,

with its Imperial capacity to defend stakeholders globally, has met its match in Asia, where China obliges itself, as a state, to follow that principle that states are dominantly driven by non-interference within the border of emerging economies. This aptly mirrors the problems surrounding achieving a single understanding of corporate purpose across the globe.

Green Industrial Policies as a Second-Best Alternative

The geopolitical landscape gives unilateral green industrial policies adopted by the predominantly industrialized countries a soft second best plausibility. The Inflation Reduction Act of the United States, the Green Deal in the EU, or India's ambitious renewable energy targets can all inspire clean technology development and large-scale deployment, even in the absence of perfect global coordination. The World Trade Organization permits environmental policy exceptions to be construed in accordance with the need of lawful measures in situations in which appropriate jurisdiction exists.

Therefore, it is a combination of these strategic green industrial policies enforced by the major economies, with climate litigation, while respecting the diversity of approaches to corporate purpose across jurisdictions, which becomes the second-best option.

The argument posed by Nurgozhayeva and Puchniak that the existing academic definitions of corporate purpose remain trapped in a narrowly jurisdictional view such that extraterritorial consequences of companies, as do formal and informal legal frameworks to address these, remain outside of their domain. Climate change and various pressing sustainability questions should be deemed unknowns of a global nature. Therefore, it warrants a view of corporate purpose beyond national borders.

But the globalization of corporate purpose is an issue without merit. The very act, then, of great powers, firms, and organizations defining the purpose of a corporation extraterritorially gives just reason for concern that corporate purpose will be defined solely for these hegemonic interests and against weaker communities. The specter of China illustrates beautifully this kind of complexity. The EU has been busy propagating a stakeholder-oriented corporate purpose worldwide, while the Chinese have taken on a more nuanced, non-interventionist position of interest in keeping with its principle of non-interference with the internal affairs of other states.

The geopolitical space offers a second-best alternative, more plausible, even when it is unilateral green industrial policies by large economies. Examples can be the intention behind the US Inflation Reduction Act or views behind EU Green Deal or India's green energy targets, inducing clean technology development and its implementation on a large scale without perfect global coordination. The WTO demand for exceptions under sufficiently specified environmental policies makes such an approach possible.

Symbiosis with this initiative focuses on findings that engage capital toward sustainable investments and accountability systems. Landmark cases globally are establishing that a government's inaction concerning climate change constitutes a human rights violation, in that there is an obligation to calculate Scope 3 emissions. Climate-washing lawsuits have won 70% of all cases.

Symbiotic relationship between green industrial policies and climate litigation:

Successful climate lawsuits, such as Milieudefensie v. Shell in the Netherlands (which required Shell to reduce its CO2 emissions by 45 percent by 2030 compared to 2019 levels), send a strong signal to companies that they must align their strategies with climate goals, which in turn will create demand in support of the clean technologies and sustainable solutions that green industrial policies emphasize.

Reasons for and limitations on dependence on unilateral green industrial policies:

Unilateral green industrial policies can lead to trade hostilities when they are considered protectionist or discriminatory by other countries. For example, the proposed EU Carbon Border Adjustment Mechanism (CBAM) that would set a carbon price on imports of certain goods to avert "carbon leakage" faces criticism from trading partners who believe it could act as a trade barrier.

Examples of green industrial policies provoking clean tech:

The U.S. Inflation Reduction Act set aside \$369 billion towards climate and clean energy, including EV tax credits, renewable energy production, and 'home retrofitting'. To boost the uptake of clean technology.

WTO Exceptions for Environmental Policies

- The maritime supply of certain provisions under GATT provide leeway for some exceptions to WTO obligations for whatever is necessary to protect human, animal, and plant life or health or that involve conservation of natural resources that are exhaustible.
- In the case of US-Shrimp (1998), this provided for a ban by the Appellate body of the WTO on shrimp caught without Turtle Excluder Devices. This was acknowledged as a conservation measure on an exhaustible natural resource.

Satellite imagery takes several urgent environmental changes into account, like rising temperatures and land degradation, from 2015-2023. Immediate action should be taken through available channels instead of waiting for a perfect global system. An interface between green industrial policies and climate litigation can create a race to the top whereby successful strategies in some jurisdictions yield policy influence in other jurisdictions, respecting the diversity of approaches to corporate purpose globally. This pragmatic "second-best" path of combining green industrial policies with climate litigation acknowledges the express lack of general agreement on, say, what should be targeted under carbon pricing but allows for progress on sustainability within the limits of market mechanisms.

I made these visualisations using Google Earth Engine to further elucidate.

DATASETS USED:

- Night Lights: NOAA/VIIRS/DNB/MONTHLY_V1/CMCFG
- Temperature: MODIS/061/MOD11A1
- Land Cover: GOOGLE/DYNAMICWORLD/V1
- Urban Analysis: LANDSAT/LC08/C02/T1_TOA

Figure 1: Night Light Intensity Changes Around Tata Steel Plant, Jamshedpur (2015-2023) [yellow to red: light intensity]

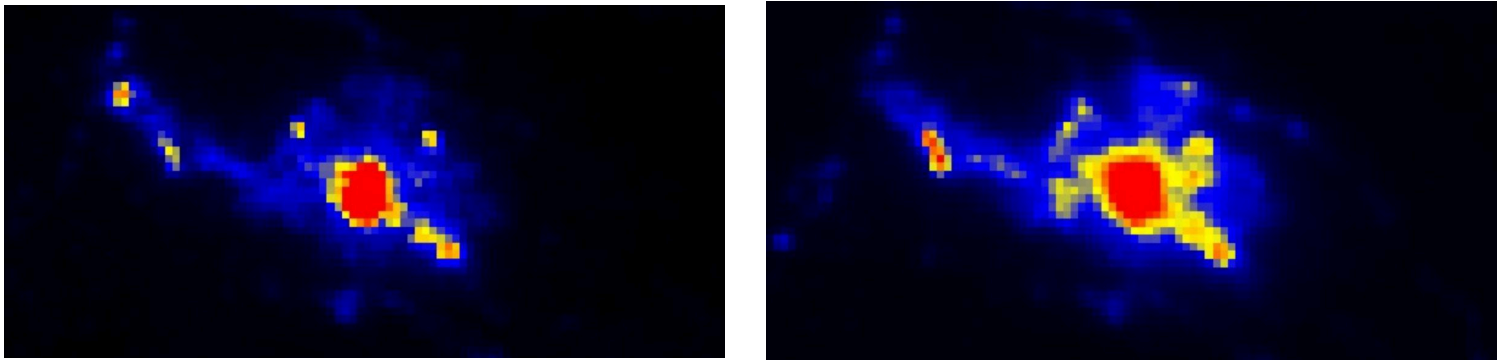


Figure 2: Surface Temperature Distribution in Jamshedpur Industrial Region (2015-2023) [Red -> hot]

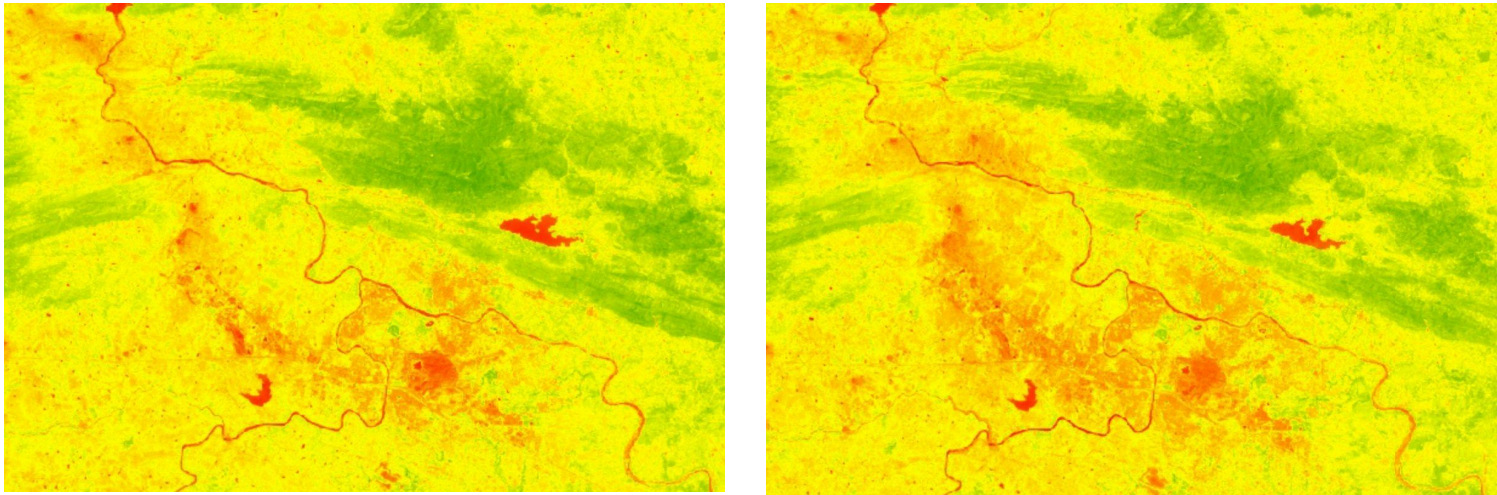


Figure 3: Land Cover Classification Map of Jamshedpur Urban Area (2015-2023) orange brown = crop land

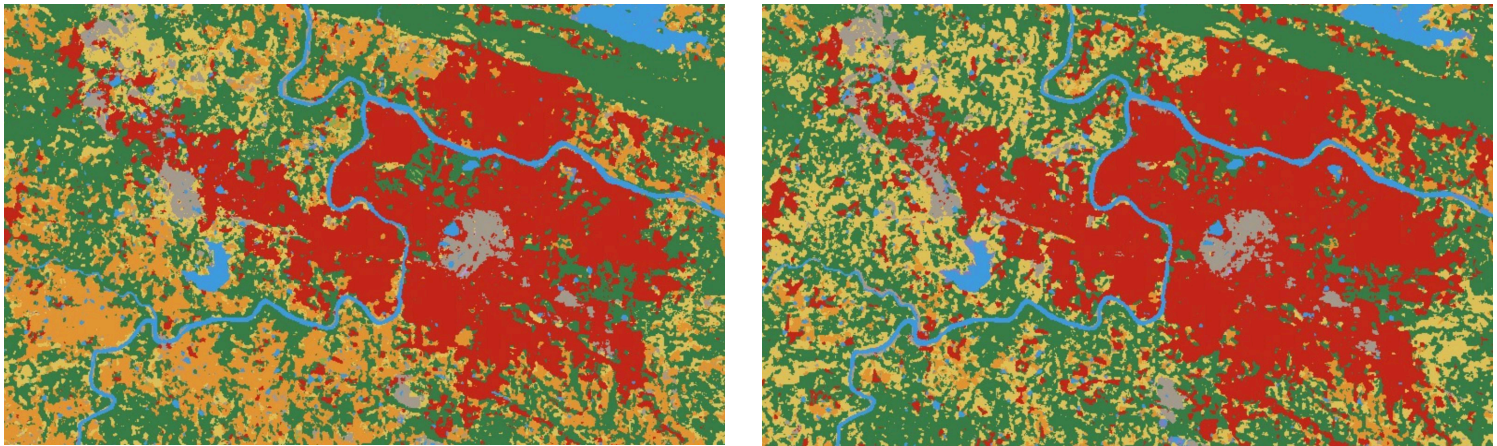
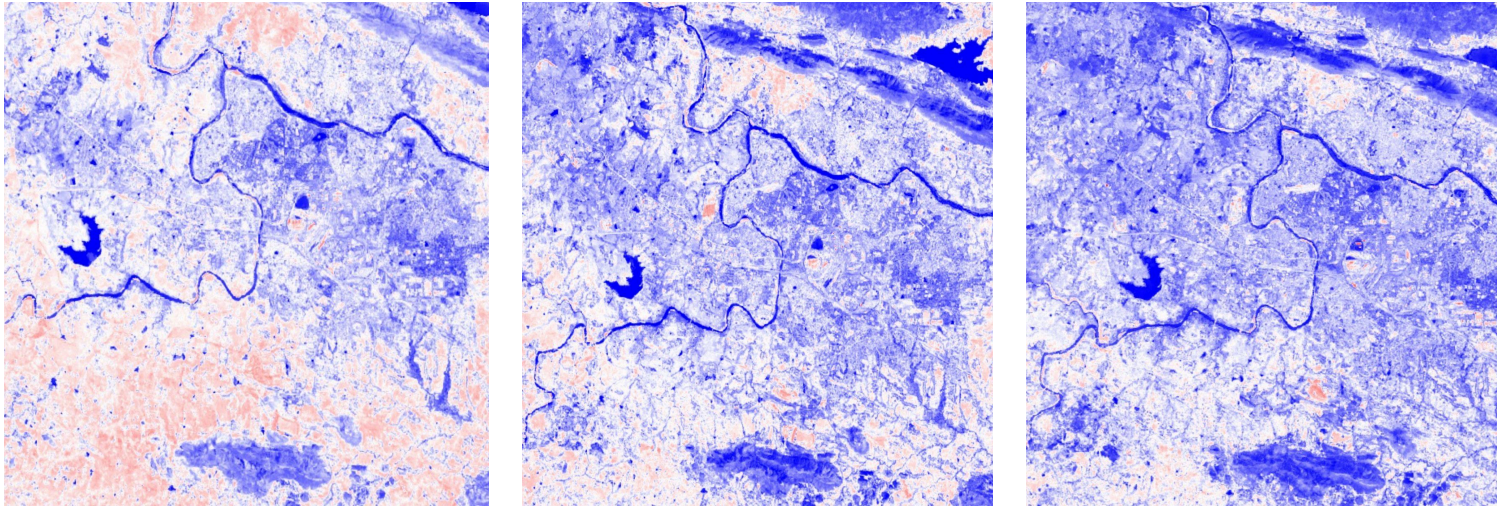


Figure 4: Urban Area Expansion Patterns in Jamshedpur Region (2015-2023)



Explaining the Figures

Fig 1: Night light imagery confirms increased illumination (red/yellow) indicating an increase in industrial activity and expansion of urbanization around the steel plant. The intensity and spatial extent of lights has grown in between the years 2015-2023.

In tandem with green industrial policies, litigation is playing a great role in the quest for corporates to meet their accountability to the environment and society and to invest sustainably. Thus far, landmark cases around the globe have established government inaction on climate change a violation of human rights and compel companies to account for and take responsibility for their Scope 3 emissions. Climate-washing lawsuits, in particular, enjoy a whopping 70% success rates, pressuring corporations into revisiting their traditional risk models into accepting a more sustainable business practice.

Fig 2: The temperatures map indicates (yellow/red colors) the formation of urban heat islands surrounding industrial zones and urban areas. There is a remarkable temperature rise, especially in and around industrial sites and extending urban regions.

This shows the rising temperature of the surface and formation of urban heat islands around industrial zones and urban areas in Jamshedpur. The colors in yellow/red call for the urgency of legal action in holding corporations accountable for their environmental impact and facilitate the establishment of green practices.

Satellite Imagery Evidence of Environmental Changes

An analysis of satellite imagery unearths the glaring environmental changes happening in Jamshedpur from 2015 to 2023. These include increases in temperature, land degradation, and rapid urbanization. All this calls for immediate actions through available political and legal channels before an ideal global system is achieved.

Fig 3: Urban land use modifications from green open space/agricultural ground to red land. Urbanization and industrial area development, justified with a thick red zone. The decline in crop fields represented by yellow-brown building is evident.

Fig 4: Analysis of built-up area stressing on the extent of urbanization away from industrial areas: darker blue indicates an increase in fair urbanization. Spreading urbanization with well-marked radii from the industrial core.

The selected images portray conversion from natural/agricultural land to built-up area, thus depicting urban sprawl and industrial growth in Jamshedpur. The yellow-brown zones and urban areas, in addition to the rapid change in red zones or the darker blues that normally suffused built-up areas, should make land use management policies and environmental impact assessment quite critical to mitigate the effects of fast land use transformation.

Statistical Analysis

In further defense of the need for urgent action I performed some basic statistical analysis on GEE data in python, from these observations of satellite-aided data can be drawn valuable insights into environmental changes that are undergoing especially in areas such as Jamshedpur.

This concerns the seasonal trend of LST (land surface temperature) data at the Tata Steel plant regarding daytime temperature variation over a significant length of time. The LST can be fitted with a sinusoidal function (Stallman, 1965) from which the mean annual LST time series, amplitude between maximum and minimum LST, and the period of oscillation can be deduced.

Time series analysis of daytime land surface temperature near the Jamshedpur TATA steel plant:

we can expect that LST has a seasonality influence and the sinusoidal trend described by [Stallman \(1965\) reading](#) as follow:

where:

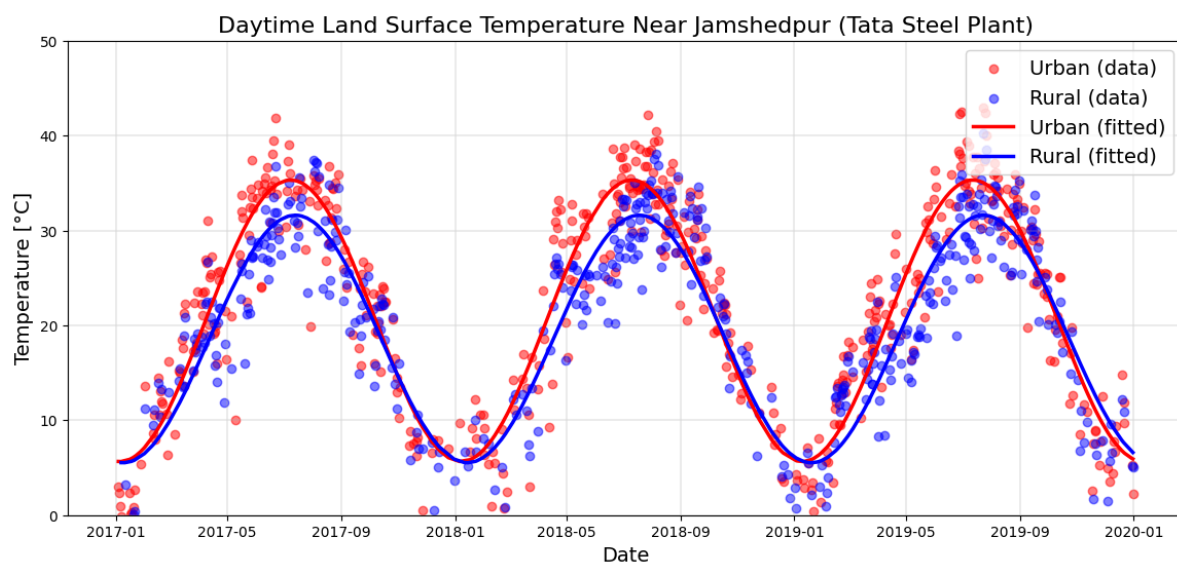
LST_0 represents the mean annual LST,

ΔLST represents the amplitude between maximal and minimal LST,

τ represents the period of oscillation of LST, and

ϕ represents an offset needed to adjust the time when $LST(t)=LST_0$

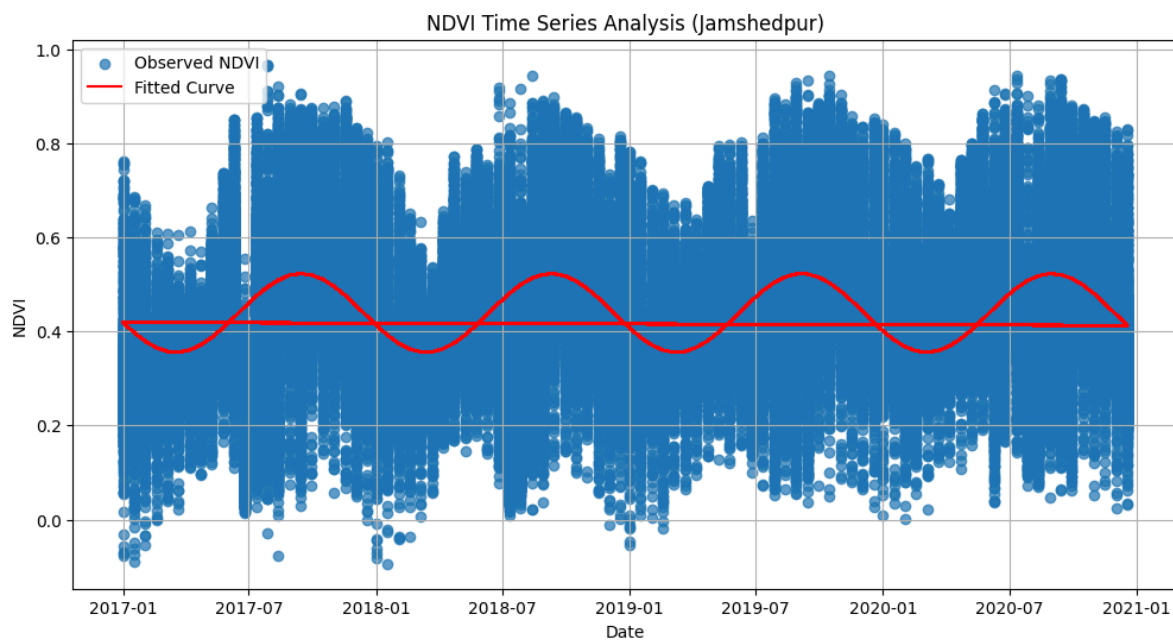
Consequently, on the top of the data scatter plot, we plot the fitting curve using the scipy library:



Do keep an eye out on the series of the NDVI: commencing from 2015-2023, against the rejected null hypothesis: significant loss or recovery trend of vegetation was obtained. [Normalized Difference Vegetation Index = (NDVI)]

Null Hypothesis: No sign of temporal trends in NDVI, for the region of Jamshedpur, 2017-2020.

Alternating Hypothesis: There exists the temporal trends of vegetation loss or recovery.



Furthermore, the Land Change in satellite imagery can be analyzed using the Chi-square test which is needed for policy and planning decisions. Findings from that will suggest the importance of land-use management by incorporating EIA and mitigation strategies to prevent the rapid transformation of a country in its land uses.

Chi Square Test: Policy and Planning Implications:

- The scale of change implicates stronger land-use management policies
- Given the significance of changes, environmental impact assessments should be a priority
- Mitigation strategies might be warranted to counter the unintended consequences of rapid land-use transformation.

Table 1: Chi-Square Test Results for Land-Use Change Analysis

Statistical Measure	Value
Chi-Square Statistic	1093.937
P-Value	7.813×10^{-231}
Degrees of Freedom	8

Table 2: 3x3 Land-Use Category Changes Across Time Periods

Land Use Category	2015	2019	2023
Natural	1491.84	749.52	665.51
Urban	14.40	32.41	8.40
Industrial	18.60	168.63	579.69

Table 3: 5x2 Land-Use Category Changes Between Time Periods

Land Use Category	2015	2023
Natural Vegetation	1491.84	994.16
Water Bodies	14.40	9.60
Agricultural Land	18.60	12.40
Built-up Area	749.52	499.48
Industrial Zone	32.41	21.59

Conclusion:

Confronted with global sustainability challenges and political obstacles towards a universal consensus of action, a pragmatic "race to the top" approach, whereby green industrial policies, climate litigation, and respect diverse corporate purpose frameworks, remain the most promising second-best alternative. Enabling the big economies to drive clean technology development, holding corporations accountable through legal action, and utilizing satellite imagery to facilitate evidence-based decisions will allow a few steps forward within the bounds of market mechanisms.

However, this second-best choice still faces numerous stumbling blocks. Political tensions and contrasting interests may disable a green industrial policy or climate litigation from achieving its objectives. Still, the present geo-economic landscape affords opportunities for action and ways by which to cultivate an environmental conscience, policymakers and corporate leaders can ensure a balance between the principles of a sustainable future. Ultimately, while definition of a corporate purpose in a universally accepted manner and perfecting a global sustainability framework seem illusive, blending strategic green industrial policies with robust climate litigation, in line with the different approaches toward corporate responsibility indicates a second-best state of affairs. Following this pragmatic route, we can go a long way toward solving global sustainability issues and crafting a more resilient and equitable future for all.

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